

Localization & Address Hierar



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Ethiopia-first localization across UI, addresses and calendar

Category: **Localization** | Locales: **EN / AM / TI** | Status: **Production**

EN / AM / TI Languages	4 levels Address levels	Greg + Eth Calendar	andegna/calender Backend pkg
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1. What it is

WaveMart is built for Ethiopia, not localized into it. The address system uses the real Ethiopian administrative hierarchy (Region / Zone / Woreda / Kebele), the UI ships in English, Amharic and Tigrinya, the calendar supports both Gregorian and Ethiopian dates, and the backend stores localized address translations in dedicated tables.

2. How it works

01 Address cascade 1	Region -> Zone -> Woreda -> Kebele, each step calls the next endpoint with the parent selected.
02 Locale selection 2	User picks EN / AM / TI; the locale is persisted in app_preferences (Hive).
03 Backend translations 3	Addresses are translated server-side; the /addresses/* endpoints accept ?locale=.
04 Date formatting 4	Dates can be rendered in Gregorian or Ethiopian; backend supports andegna/calender.
05 Tigrinya fallback 5	Tigrinya (ti) is mapped to Amharic Material localizations for missing widgets.

3. Capabilities

Capability	What it does
Multi-language UI	Full ARB-based translations in EN, AM, TI; flutter gen-l10n runs on build.
Address hierarchy	Cascading dropdowns driven by the localized DB tables.
Locale-aware formatting	Numbers, currency and dates follow the active locale.

Ethiopian calendar	Both calendar systems are supported, with side-by-side display option.
Address resolution	When a user saves an address, its ID is stored on the listing / order for analytics.
Backend translations	DB-backed translation loader (translations table) with per-locale caching.

4. Investor relevance

WHY IT MATTERS

Deep localization is a barrier to entry for foreign competitors and a force multiplier for adoption. Every Region / Zone / Woreda / Kebele we capture improves search relevance, recommendations and future mortgage-product matching.

5. Tech surface

Component	Location
Localization (mobile)	lib/l10n/app_en.arb, app_am.arb, app_ti.arb
Address service	lib/data/services/address_service.dart
Date package	kenat (Gregorian + Ethiopian)
Backend locales	WaveMart/lang/ + DB translations table
Address migrations	WaveMart/database/migrations/2026_01_19_000000_create_localized_addresses_tables.php
Calendar helpers	WaveMart/app/Helpers/EthiopianDateHelper.php + @ethiopianDate directive

6. API

```
GET /api/addresses/regions?locale=am
GET /api/addresses/zones?region=Tigray&locale;=am
GET /api/addresses/woredas?region=Tigray&zone;=Central&locale;=am
GET /api/addresses/kebeles?region=Tigray&zone;=Central&woreda;=01&locale;=am
```